



FIRST SEMESTER 2026-2027

Course Handout Part II

Date: 31 July 2026

In addition to Part-I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course Code : DE G531
Name of the course : Product Design
Instructor-in-charge : Radha Raman Mishra

1. Course Description:

The course covers various approaches to designing and developing a successful product. User research, requirements analysis, problem-specific creative solution-finding, costing, synthesis and analysis of design are included. Modularization necessary for administering large design exercises and design decision-making processes, is deliberated. Concepts of value engineering, Engineering ethics, and Design for X, along with case studies, are also incorporated.

2. Scope and Objective of the course:

This course is designed to impart the knowledge required to develop a new product – understand the opportunity, develop a concept, and implement a concept. After the successful completion of this course, students shall be able to understand and use the various processes, tools and techniques required for designing a product: product specification development; concept generation, concept selection, concept testing and embodiment; product architecture; industrial design; design for X; analytical and numerical models; physical prototypes, models and experimentation; environmental, economic and social issues in product development; patents and intellectual properties.

3. Textbook:

- T1. Kevin Otto and Kristin Wood, “Product Design”, Pearson Education, 2001, Paperback Indian Ed-2013
T2. Ulrich, K. T., Eppinger, S. D., and Yang, M. C. Product design and development, 7th Edition, McGraw-Hill Higher Education, 2008.

4. Reference Book:

- R1. David G. Ullman, “The Mechanical Design Process”, 4th Edition, McGraw-Hill Higher Education, 2009.
R2. N. J. M. Roozenburg, J. Eekels, N. F. M. Roozenburg, "Product Design: Fundamentals and Methods", John Wiley and Sons, 1995.

5. Course Contents

L. No.	Topics	Learning Objectives	Book (Chapter)
1-4	Introduction	Introduction to design, modern product development process, and innovative thinking	T1 (1)





5-6	Product Development Process and Product Planning	Product development teams, Product development planning process and basic tools	T1 (2)
7-8	Scoping Product Development	What to develop? Mission statement, Technical questioning, Business case analysis, Design drivers	T1 (3)
9-16	Product Specification Development	Identifying the customer needs, establishing product function, Benchmarking, and establishing product specification	T1 (4, 5, 6, 7)
17-18	Product Architecture	Architecture types, modularity design, implications	T1 (9)
19-23	Conceptual Design	Concept generation, concept evaluation, selection, testing and embodiment	T1 (10, 11, 12)
24	Prototyping	Types of prototypes, Uses of Prototypes, Techniques	T1 (17)
25-27	Industrial Design	Need for industrial design, industrial design process, management and assessing the quality of industrial design	T2 (10)
28-31	Design for Manufacturing and Assembly	Methods of designing for manufacturing and assembly	T1 (14)
32-35	Design for Environment	Methods of designing environmentally conscious products	T1 (15)
36-37	Design for Robustness	Taguchi Method: Parameter design, process design, S/N ratio	T1 (19)
38-40	Product Development Economics	Economic analysis process, qualitative and quantitative analyses	T1 (15)

6. Evaluation Scheme and Schedule:

S. No.	Evaluation Component	Duration	Weightage (100%)	Date & Time	Remarks
1	Mid-semester Examination	90 minutes	25	10/10 AN2	CB/OB
2	Quiz		10	To be announced in class	
3	Project/Assignment		30	To be announced in class	
4	Comprehensive Examination	180 minutes	35	16/12 AN	CB/OB

7. Chamber Consultation Hour: Thursday, 12 pm- 1 pm.

8. Notices concerning the course: All notices concerning the course will be communicated via email/ Google Classroom.

9. Make-up Policy: Make-up will be permitted only in genuine cases with prior permission. Other than medical reasons, prior permission of the IC is compulsory. No make-up for quizzes.

Instructor-in-Charge

DE G531

